AegisLink

Console-Based Encrypted File & Message Transfer System

Software Project Lab

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Table of Contents

01 — Project Description

02 — Motivation

03 — Working Methodology

04 — Technology Used

05 — How DES Works

06 — How AES Works

07 — How SHA-512 Works

08 — Progress

Project Description

AegisLink enables secure file and message transfer between clients over a TCP network. All data is encrypted before transmission and verified for integrity on arrival.

Encryption	Integrity	Networking
DES and AES algorithms encrypt all messages and files before sending.	SHA-512 hashing detects tampering — server flags [INTACT] or [TAMPERED!].	Multithreaded TCP server on port 314159 handles multiple clients at once.

Motivation

01

Security

Data sent over a network can be intercepted. DES and AES encryption ensures only the intended receiver can read it.

02

Integrity

Data can be altered in transit. SHA-512 hashing lets the server verify every message and file arrived unmodified.

03

Usability

Console-based and lightweight — no complex setup needed for either the client or server.

Working Methodology

01

Connection

Client connects to server via TCP (port 314159), enters name, selects DES or AES.

02

Encryption

Message or file is encrypted locally before leaving the client.

03

Hashing

SHA-512 hash of the ciphertext is computed and attached to the data.

04

Transmission

Encrypted data + hash is sent over the TCP socket to the server.

05

Verification

Server recomputes hash and compares — flags [INTACT] or [TAMPERED!].

06

Relay

If intact, server decrypts and forwards to all other connected clients.

Technology Used

DES — Data Encryption Standard

- 64-bit block, 56-bit key, 16 rounds
- Feistel network structure
- Legacy — included for comparison

AES — Advanced Encryption Standard

- 128-bit block, 128-bit key, 10 rounds
- SubBytes → ShiftRows → MixColumns
- Current industry standard

SHA-512 — Hashing

- Produces 512-bit digest of ciphertext
- 1-bit change → completely different hash
- Flags [INTACT] or [TAMPERED!]

TCP Socket Programming

- Winsock2, port 314159
- std::thread per connected client
- Supports messages and file transfer

How DES Works

01 The 64-bit plaintext block goes through an Initial Permutation (IP) to scramble bit positions.

02 The block is split into Left (L) and Right (R) halves of 32 bits each.

03 16 Feistel rounds run: R is expanded to 48 bits, XORed with the round key, passed through S-Boxes, then permuted.

04 After each round: new R = old L XOR f(R, Key). New L = old R.

05 After 16 rounds, halves are swapped and a Final Permutation (FP) produces the 64-bit ciphertext.

Key size: 56 bits · Block size: 64 bits · 16 rounds

How AES Works

01 The 128-bit plaintext is arranged into a 4×4 byte matrix called the State.

02 AddRoundKey: The initial round key is XORed with the State before any rounds begin.

03 SubBytes: Each byte is replaced using a fixed lookup table (S-Box) for non-linearity.

04 ShiftRows: Row 0 stays, Row 1 shifts left by 1, Row 2 by 2, Row 3 by 3.

05 MixColumns: Each column is multiplied in $GF(2^8)$ to spread bits across the block. (Skipped in the final round.)

Key size: 128 bits · Block size: 128 bits · 10 rounds · Each round: SubBytes → ShiftRows → MixColumns → AddRoundKey

How SHA-512 Works

01 The message (ciphertext) is padded so its length is a multiple of 1024 bits, then split into 1024-bit blocks.

02 8 fixed hash values (H0–H7) are initialized — these are derived from the first 8 prime numbers.

03 Each 1024-bit block goes through 80 rounds of mixing using bitwise operations (AND, XOR, rotate) and 80 round constants.

04 After each block, the result is added to the previous hash values. This chaining ensures every bit of input affects the output.

05 The final 512-bit (64-byte) digest is produced. Any change in the input — even 1 bit — produces a completely different hash.

Output: 512-bit digest · Used in AegisLink to verify every message and file arrives unmodified

Progress

Done

TCP Server Implementation

Multithreaded, handles multiple clients

Done

TCP Client Implementation

Connect, send messages and files

Done

Client-Server Communication

Bidirectional message relay confirmed

Done

DES Algorithm

Encrypt / decrypt messages and files

Done

AES Algorithm

Encrypt / decrypt messages and files

Done

SHA-512 Integrity Verification

Hash check on every transfer

Output — Encrypted Chat (DES)

Client 1 — meem

```
C:\Users\Jannatul Meen
--- ENCRYPTED CHAT (DES & AES) ---
Connected to server!

Enter your name: meem
Choose encryption (1=DES / 2=AES): 1

[Chat ready | meem | DES]
  /file <path> - send a file
  /quit        - exit
----------------------------------------
> halo. ich bin meem.
>
>> nabila : halo. ich heiBe maria. wie geht's
?
> sehr gut. und du?
>
>> nabila : mir geht es gut.
> freut mich.
>
```

Client 2 — nabila

```
C:\Users\Jannatul Meen
--- ENCRYPTED CHAT (DES & AES) ---
Connected to server!

Enter your name: nabila
Choose encryption (1=DES / 2=AES): 1

[Chat ready | nabila | DES]
  /file <path> - send a file
  /quit        - exit
----------------------------------------
>
>> meem : halo. ich bin meem.
> halo. ich heiBe maria. wie geht's?
>
>> meem : sehr gut. und du?
> mir geht es gut.
>
>> meem : freut mich.
>
```

Server

```
C:\Users\Jannatul Meen
   ENCRYPTED SERVER - DES & AES Support
Server started on port: 314159
Supports DES and AES encryption
Waiting for clients...

Client connected.
Client connected.
[INTACT][DES] meem : halo. ich bin meem.
[INTACT][DES] nabila : halo. ich heiBe maria.
 wie geht's?
[INTACT][DES] meem : sehr gut. und du?
[INTACT][DES] nabila : mir geht es gut.
[INTACT][DES] meem : freut mich.
```

Server logs show [INTACT][DES] — confirming messages arrived unmodified.

Output — File Transfer (AES) + Integrity Check

Client — sending file.txt via AES

```
C:\Users\Jannatul Meen
--- ENCRYPTED CHAT (DES & AES) ---
Connected to server!

Enter your name: meem
Choose encryption (1=DES / 2=AES): 2

[Chat ready | meem | AES]
  /file <path> - send a file
  /quit        - exit
----------------------------------------
> /file C:\Users\Jannatul Meem\file.txt
[File sent: file.txt | AES]
>
```

Server — receiving, hash check, save

```
C:\Users\Jannatul Meen
   ENCRYPTED SERVER - DES & AES Support
Server started on port: 314159
Supports DES and AES encryption
Waiting for clients...

Client connected.
[INTACT][DES] meem : quit
Client Disconnected!
Client connected.
Client Disconnected!
Client connected.
Receiving encrypted file: file.txt (272 bytes
) [AES]
[TAMPERED!] File integrity check failed: file
.txt
File "file.txt" saved successfully! [AES]
```

Server detects [TAMPERED!] on hash mismatch, then saves the file successfully with AES decryption.

Thank You

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GitHub Repository

https://github.com/Jfmeem/AegisLink